Course Code: CSE3714	**Course Name:** Network Security	
Credits: 3 (3-0-0)	**Contact Hours: 3** hour Theory per week	
Batch: 2023, 5th Sem **Academic Year:** 2025-26	**Semester Duration: 29th** Aug 2025 to 6th Dec	
Course Faculty: Dr. Satyendr Singh Contact: +91-8683856238 Email: satyendr.singh@bmu.edu.in Office: 80, IV Floor, E2 Building Dr. Pramod Kumar Maurya Contact: +91-9643322724 Email: pramod.maurya@bmu.edu.in Office: 7, IV Floor, E2 Building	**Course Coordinator:** **Name:** Dr. Satyendr Singh **Email:** satyendr.singh@bmu.edu.in **Office:** 80, IV Floor, E2 Building	

Aim of the course: The goal of this course is to understand the principles and practice of cryptography algorithms and fundamentals of security protocols and techniques. It covers cryptography, network-based security threats and vulnerabilities, and practical solutions to system security, IP Security, Web Security, Network Management Security and System Security.

Course Overview and Context: Network Security is a V semester core course in the undergraduate program in Computer Science and Engineering. The course focuses on the principles and applications of cryptographic techniques, in order to provide the basis for important aspects such as network and data security. It covers Network Security, IP Security, Web Security, Network Management Security and System Security.

Topics of the Course

- Introduction: Attack, Services and Mechanism, Model for Internetwork Security.
- Cryptography, Notion of Plain Text, Encryption, Key, Cipher Text, Decryption and Cryptanalysis, Public Key Encryption
- Hash function, MAC
- Digital Signatures.
- IP Security Architecture: Overview, Authentication header, Encapsulating Security Pay Load combining Security Associations, Key Management.
- Transport Layer Security: Secure Sockets Layer, Transport Layer Security.
- Electronic Mail Security: Pretty Good Privacy (PGP), S/ MIME
- System Security: Intruders, Viruses and related threads, Firewall design principles.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand cryptography and network security concepts with their applications	**Employability**	1. Assessment on cryptography and network security basics 2. Discuss Real-world applications of cryptographic techniques 3. Analyze a security breach

			and explain cryptographic weaknesses
CO2	Apply security principles to design a variety of secure systems	**Skill Development**	1. Implement symmetric and asymmetric encryption using programming languages 2. Apply CIA (Confidentiality, Integrity, Availability) principles to a system design
CO3	Analyze and design network security protocols	**Entrepreneurship, skill Development**	Compare existing security protocols (e.g., SSL/TLS, IPSec) and propose improvements

Prerequisite: if any

1. Computer Networks

Topics of the course:

Topics	Number of sessions	Course Outcomes
Cryptography and basic terminologies, Security goals, Security attacks-Active and Passive, Steganography, Services and Mechanisms for Internetwork Security.	4	CO1
Conventional encryption model, Cryptanalysis, Symmetric and Asymm Encipherment, Classical encryption techniques-substitution ciphers an transposition ciphers, Stream and block ciphers.	6	CO2
Modern Block Ciphers, Block cipher principles, Block cipher modes of operations.	3	CO1
Data Encryption standard (DES), Strength of DES, Triple DES.	3	CO2
Principals of public key cryptosystems, Application and requirements, RSA algorithm, Security of RSA Key management, Understand and apply the Diffie- Hellman key exchange algorithm	5	CO3
Digital signature, RSA Based digital signature, Message authentication code, Hash function, Security of hash functions and MACS, Secure hash algorithm (SHA).	6	CO1
IP Security: Overview, Authentication header, Encapsulating Security Pay Load combining Security Associations, Key Management	5	CO3
Transport Layer Security: Web Security Issues, Secure Sockets Layer (SSL), Transport Layer Security (TLS).	5	CO3

Electronic Mail Security: Pretty Good Privacy (PGP), S/ MIME	5	CO3
System Security: Intruders, Viruses and related threads, Firewall design principles.	3	CO3

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	2	1													3	3	
CO2	2	1			2										3	3	
CO3	2	1			2										3	3	

Learning Resources:

Textbook:

1. Atul Kahate, "Cryptography and Network Security", Tata McGraw Hill Education, 4e, 2019.
2. Behrouz A Forouzan & Debdeep Mukhopadhyay, "Cryptography and Network security", 3rd Edition, McGraw Hill, 2015.
3. William Stallings, "Cryptography and Network Security", 7th Edition, Prentice-Hall / Pearson Education, Englewood Cliffs /New Delhi, 2017.
4. Willian Stallings, "Network Security Essentials, Applications and Standards", 6th Edition, Pearson Education.

Refernce Books:

1. Michael Goodrich & Roberto Tamassia, "Introduction to Computer Security", Prentice-Hall/ Pearson Education, Englewood Cliffs/ New Delhi, 2010.

Video lectures/ MOOC:

1. MIT equivalent course (Network and Computer Security): https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-857-network-and-computer-security-spring-2014/index.htm

2. NPTEL equivalent course (Cryptography and Network Security): http://nptel.ac.in/courses/106105031/

Experiential Learning Component:
The Experiential Learning Component of the Web Programming course (CSE2716) is designed to provide students with hands-on experiences that reinforce theoretical knowledge through practical application. Students will be encouraged to code the ciphers and other aspects in Network Security in Python.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the internal assessment evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics
Assignment	20%	Continuous	
Mid Sem	20%	As per academic calendar/Date-sheet	Close book written exam
Quiz	20%	November 2nd week	Topics to be covered will be announced in the class. 20 MCQ questions, each of 1 mark. The quiz will be offline.
End Term Evaluation	40%	As per academic calendar/Date-sheet	Close book written exam

- All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code.
- Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity.
- There is a mandatory requirement to upload the project to a public repository on GitHub.

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.